

Engaging the Patient in Healthcare:
Conversational Assistants
in the era of LLMs

BARBARA DI EUGENIO
UNIVERSITY OF ILLINOIS CHICAGO
CHICAGO, IL, USA



The bias that blinds: why some people get dangerously different medical care

Medical research and practice have long assumed a narrow definition of the 'default' human, badly compromising the care of anyone outside that category. How can this be fixed?

by [Jessica Nordell](#)

LOWN RIGHT CARE
Reducing Overuse and Underuse

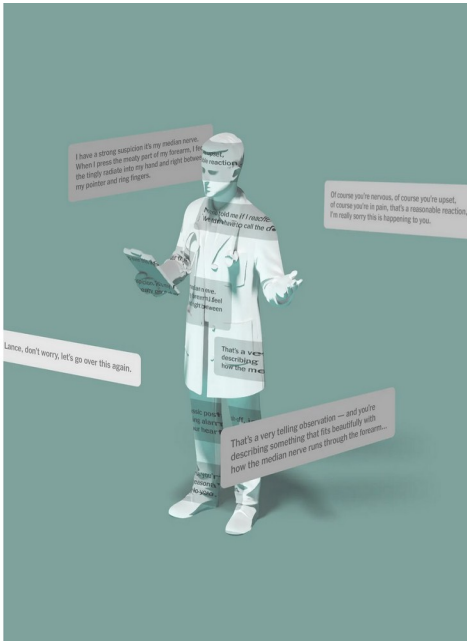
Poor Physician-Patient Communication and Medical Error

PDF Print Comments

ANDY LAZRIS, MD, CMD, Personal Physician Care, Columbia, Maryland
ALAN R. ROTH, DQ, FAAP, FAAHPM, Jamaica Hospital Medical Center, Jamaica, New York
HELEN HASKELL AND JOHN JAMES
Am Fam Physician. 2021;103(12):757-759

Frustrated by the Medical System, Patients Turn to A.I.

Chatbots are cheap, always available, superficially empathetic — and sometimes wrong. Some have concluded they're a risk worth taking.



Improving Patient-Provider Communication A Call to Action

Patak, Lance MD, MBA, RN; Wilson-Stronks, Amy MPP, CPHQ; Costello, John MA, CCC-SLP; Kleinpell, Ruth M. PhD, RN, FAAN; Henneman, Elizabeth A. PhD, RN; Person, Colleen MMA, BSN, RN; Happ, Mary Beth PhD, RN

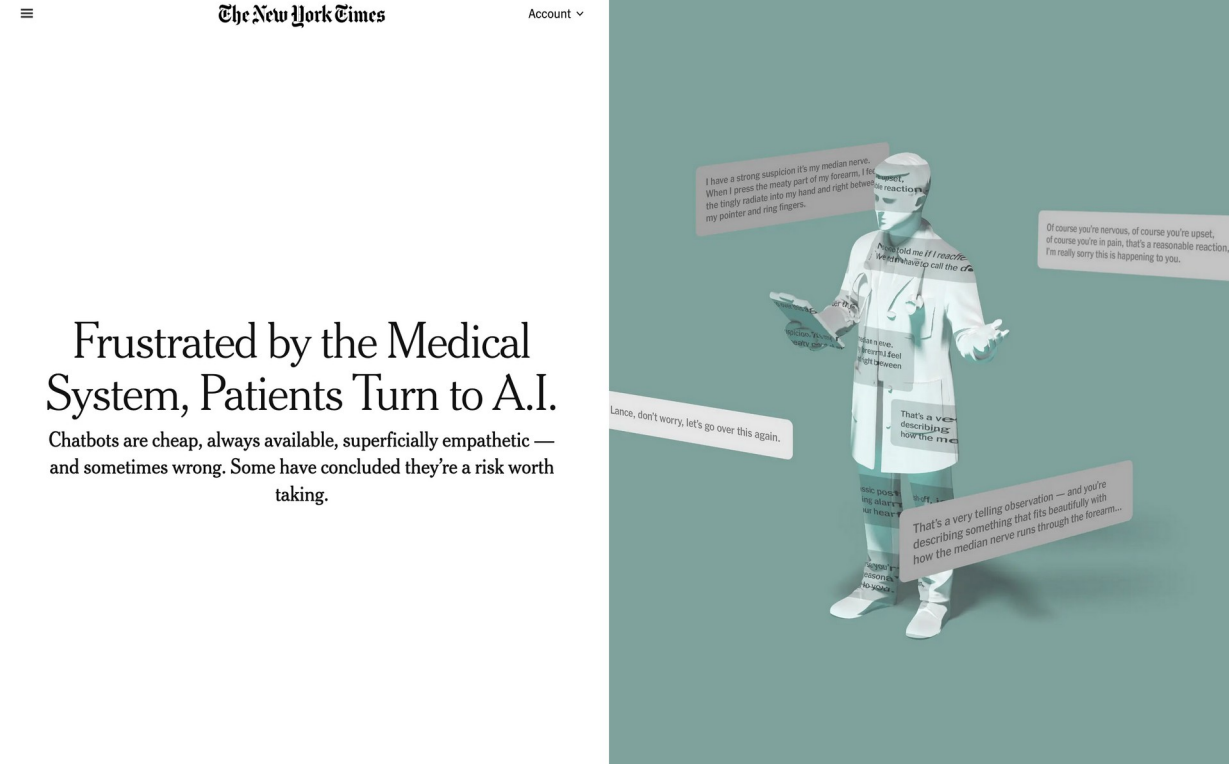
Author Information

JONA: The Journal of Nursing Administration 39(9):p 372-376, September 2009. | DOI: 10.1097/NNA.0b013e3181b414ca



Conversational assistants for patients

Hypothesis: Addressing patients' needs when accessing data / communicating with healthcare providers → better health outcomes



Conversational assistants for patients: A non-exhaustive list of challenges

Challenges:

- ❖ **Datasets:** ecologically valid data. Often no data / small data / data not sharable
- ❖ **Computational models** (AI/ML/NLP) must be:
 - ❖ informed by interdisciplinary insight: linguistics, psychology, health sciences
 - ❖ explanatory (no black-box models)
 - ❖ fair and not biased

Conversational assistants for patients: A non-exhaustive list of challenges (ctd)

❖ Computational models:

- ❖ **LLMs** (Large Language Models) are not there yet ... if they will ever be:
 - Hallucinations, biases, opaqueness. How can we use them appropriately?
 - Are **neuro-symbolic** approaches the / one answer?

- ❖ **Evaluation**: no SOTA (state-of-the-art) to beat; measure impact on real users / patients

Challenges Incarnated

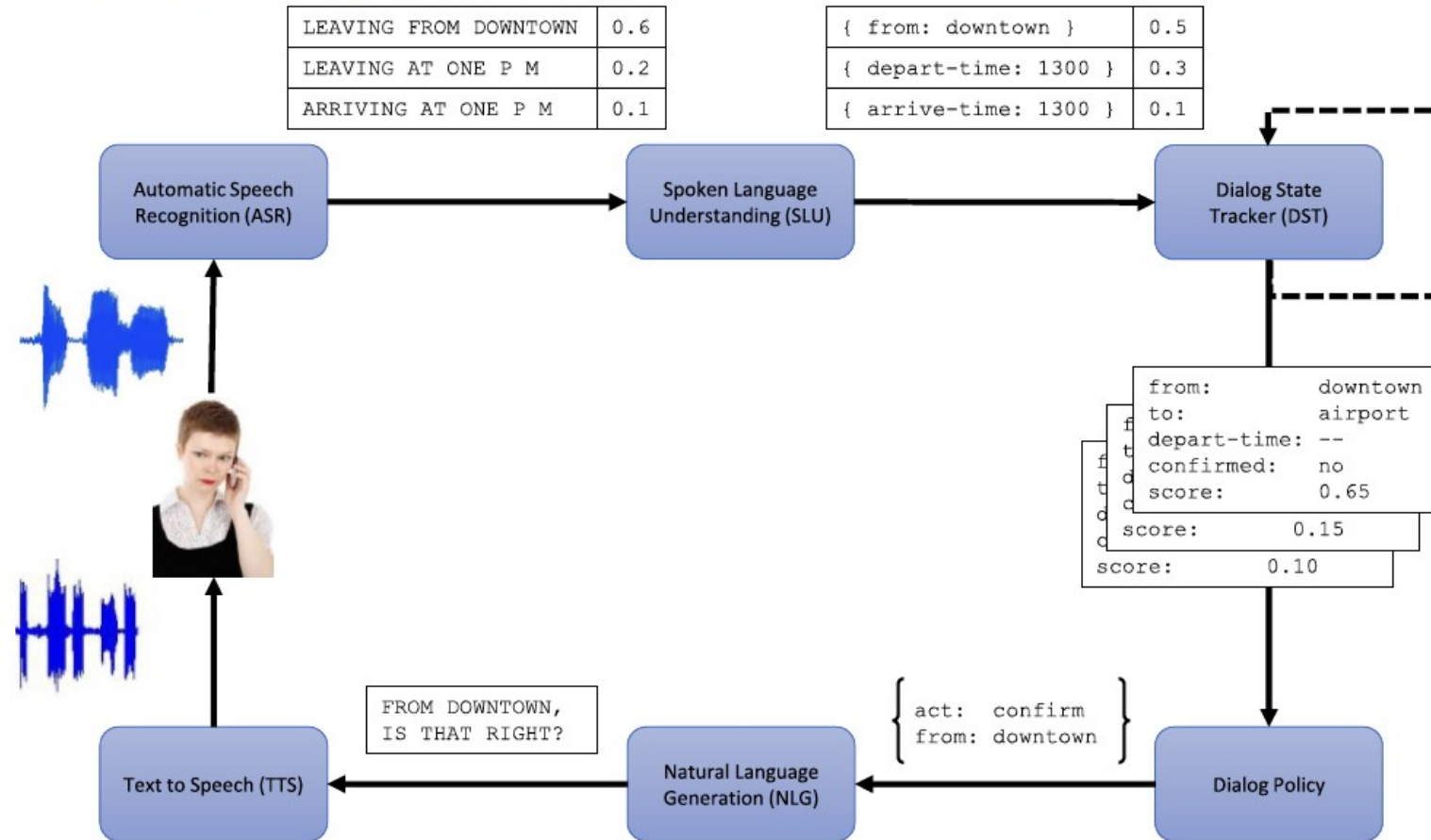
- ❖ Virtual Health Coach: support patients (and human coaches) in adopting SMART goals via SMS exchanges
- ❖ HFChat: conversational agent for Heart Failure patient education

Riassunto- lezione 7 su dialogue systems & chatbots (Prof Mazzei)

6 Key features of human dialogues

1. Turns
2. **Speech acts**
3. Grounding
4. Dialogue structure
5. Initiative
6. Implicature

DS Architecture



Chatbots: how do they answer?

Corpus based chatbots: 2 ways

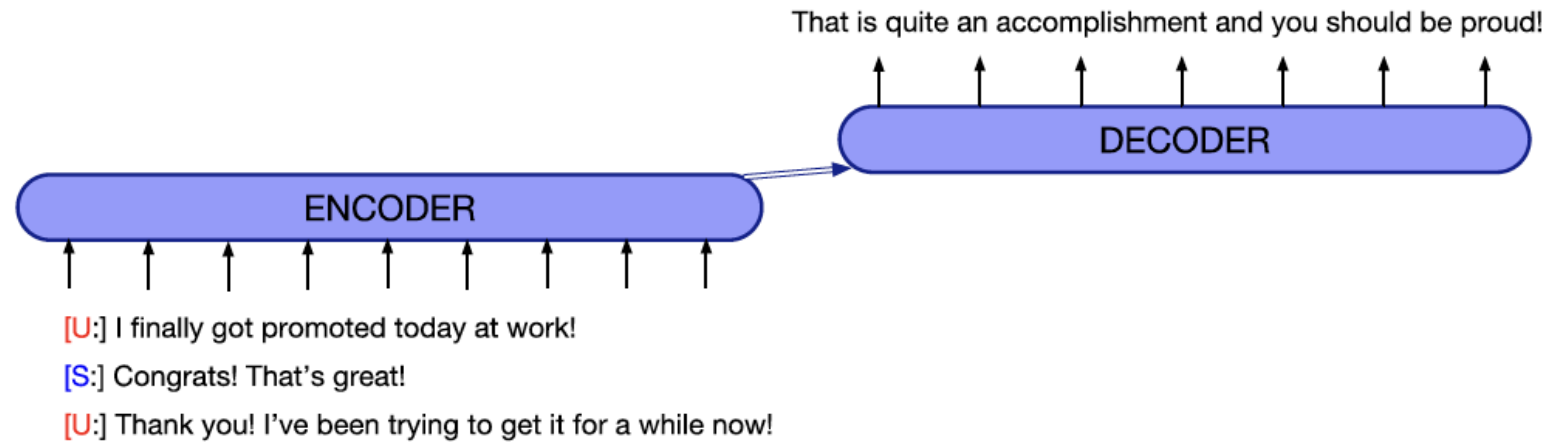
1. Response by retrieval

- Use information retrieval to grab a response (that is appropriate to the context) from some corpus

2. Response by generation

- Use a language model or encoder-decoder to generate the response given the dialogue context

Response by generation

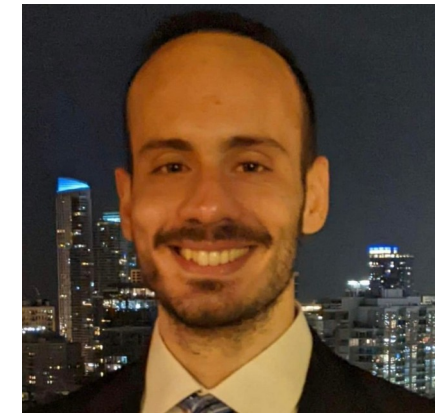


Virtual Health Coach (National Science Foundation 1838770, 2018-2025)

4 collaborators, 2 institutions, 2 colleges



3 PhD students



Health Coaching

Health coach serves as advisor that provides ongoing support to clients as they set goals and make sustainable changes to improve their health [Kivelä et al., 2014]

Limitations of health coaching:

- Expensive
- Time-intensive
- Inflexible
- Limited reach based on travel
- Waning interest with time

Virtual Coach: Overall Research Question

How do we motivate people to adopt physically active lifestyle and help them maintain it long term in cost-effective way?

❖ Dialogue-based **virtual assistant health coach** to interact via text messages and help increase physical activity by setting **S.M.A.R.T. goals** (Doran, 1981).

I will **walk on my lunch break** for **20 mins Wed and Thurs this week**

Specific / Measurable / Attainable / Realistic / Time-bound

❖ Intermediate step: **support health coaches**

Data Collection

De-identified
dataset 2
available
<https://github.com/uic-nlp-lab/virtualcoachdata>

	Dataset 1	Dataset 2
# health coaches	1	3
# patients	30 (28)	28 (27)
Duration	4 weeks	8 weeks
Total # messages	2853	4134
Coach/Patient messages	54/46%	58/42%
Activity tracker	Fitbit	
Communication	MyTapp and SMS	
Goal setting	Weekly SMART goals	

What should dialogue agent do [to support coaches]?

Coach: Hi, hope you're well. I forgot to message you yesterday to set your goal for Week 6, my apologies!

Coach: You met your goal 5 out of 7 days last week, great job What would you like to do for this week?

Patient: Gd morning, I am going to set the same goals. With a 7 confidence.

Dialogue agent should:

- **Summarize Goal:** [activity] [location] [days] [time] ...
- **Suggest** next message to human coach

Text Messages Exchange Excerpt

Stage: Goal Setting

SMART Attributes: blue

-----Phase: Goal Identification-----

Coach: Think about this week, let's call it week 1 of 4. Now what goal could you make that would allow you to do more walking (activity)?

Patient: I can take the stairs (activity) at work (location) for the work week.

-----Phase: Goal Refining-----

Coach: Sounds good. So will this be for coming, going and at lunch time (time)?

Patient: I will do at least twice (repetition) during workdays (days).

-----Phase: Anticipate Barrier-----

Coach: On a scale of 1-10 with 10 being very sure and 1 not at all sure. How sure are you that you will accomplish your goal?

Patient: 9 (score)

Three-Tier Dialogue Annotations

- ❖ Stages and Phases ($\kappa = 0.93$)
 - higher-level portions that capture different tasks and sub-tasks
- ❖ Dialogue Acts
 - utterance level information to understand intent behind sender's message in context
 - subset of iso 24617-2 standard DAs [Bunt et al 2017] (validated, hence no κ)
- ❖ SMART Attributes ($0.51 < \kappa < 0.89$ at word level)
 - domain-specific slot values to track state of patient's goal

Interface with Summary

mytapp | Sample Study

UsersTemplatesSubjectsGroupsCohortsSurveysProgramsSensorsStudiesMy Account

[Patient name here]

Details

Groups 0

Programs 0

Sensors 0

Message:

Compose message...

1600 characters remaining

Template:

Survey:

No survey

Suggested Message:

Activity	Time	Location	Amount	Distance	Duration	Days name	Days num	Repetition	Attainability Score
walk	between 6 am to 8 am		4,000 steps	2 miles	30 min	friday saturday		a day	10

Messages

Great! let's make the first SMART goal for your first week.
Sent Monday, Jul 01, 2020 at 12:05 PM-12

I want to walk 30 min a day between 6am to 8am

Sent Monday, Jul 01, 2020 at 12:08 PM-12

Okay. We can start tomorrow morning. Would you like to do that every day or only until Friday? Saturday?

Sent Monday, Jul 01, 2020 at 12:09 PM-12

I want to start today

Sent Monday, Jul 01, 2020 at 12:10 PM-12

2,000 steps is 1 mile, 4,000 steps is 2 miles...is there a specific number of steps you would like to walk today and then in the morning starting tomorrow?

Sent Monday, Jul 01, 2020 at 12:11 PM-12

Okay for today I'm going to make 4000steps

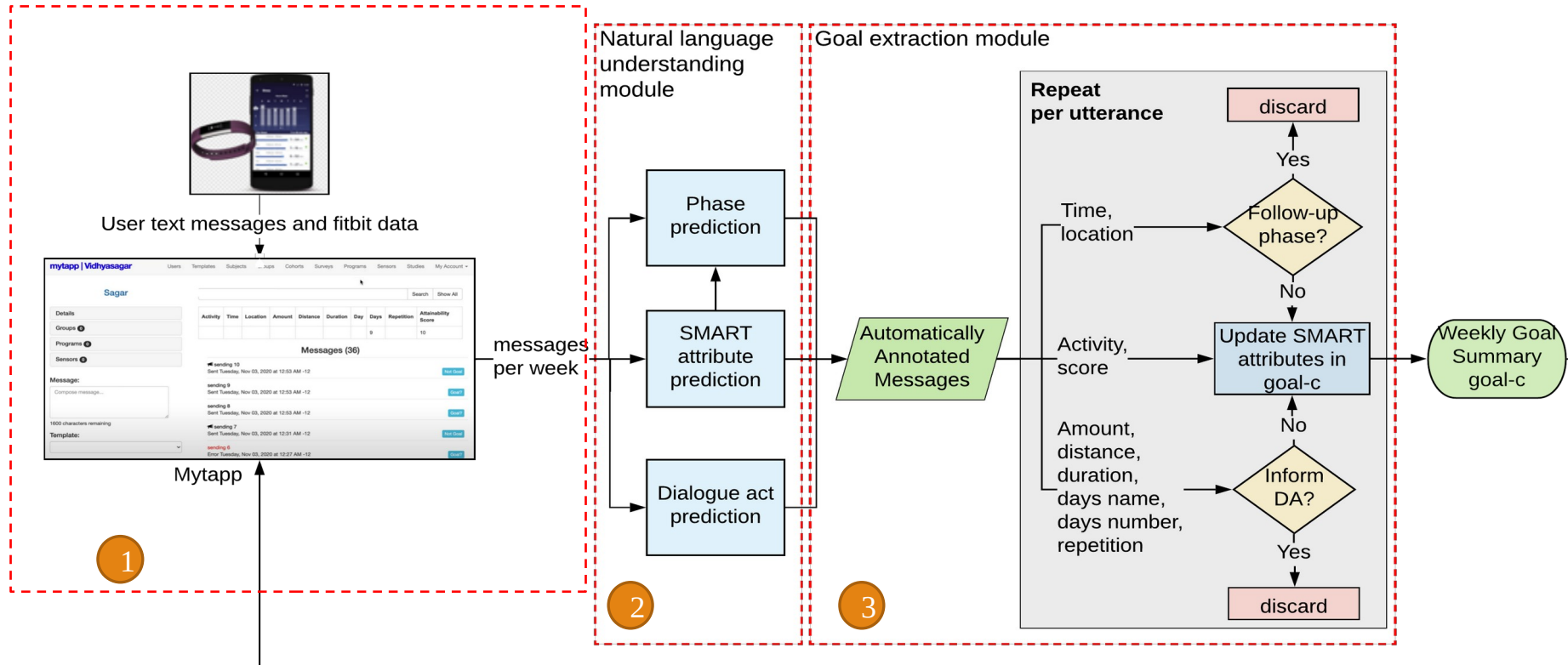
Sent Monday, Jul 01, 2020 at 12:14 PM-12

Okay, so I just want to be clear: You want to walk 4,000 steps during the morning (from 6am - 8am) this week

Sent Monday, Jul 01, 2020 at 12:15 PM-12

"Classic": Goal Extraction Approaches

[Gupta et al SigDIAL 20, 21]



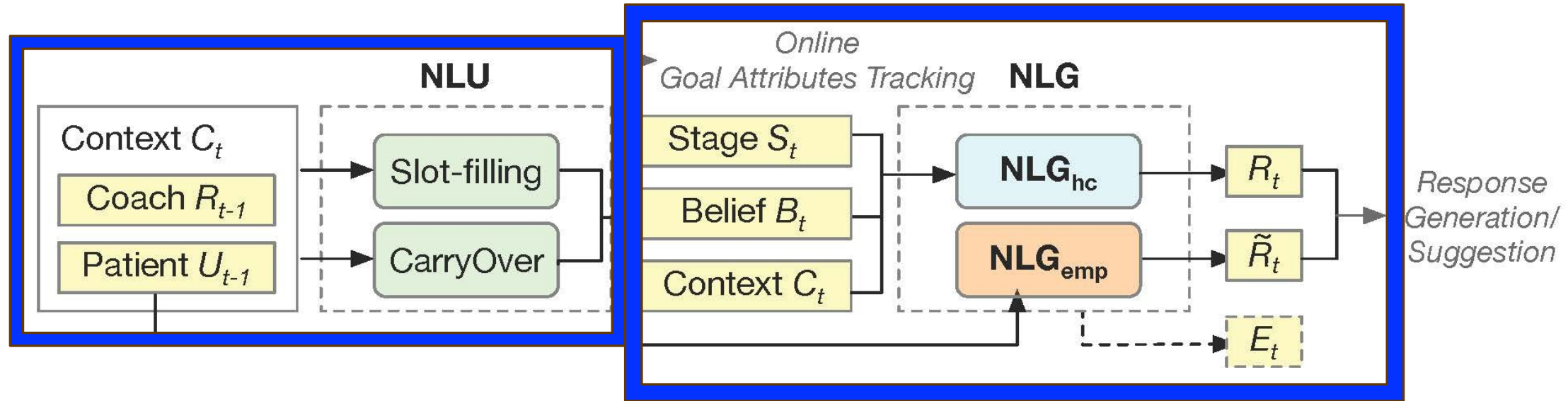
“Classic”: Goal Extraction Approaches [Gupta et al SigDIAL 20, 21]

Summarized goal with its 10 attributes:

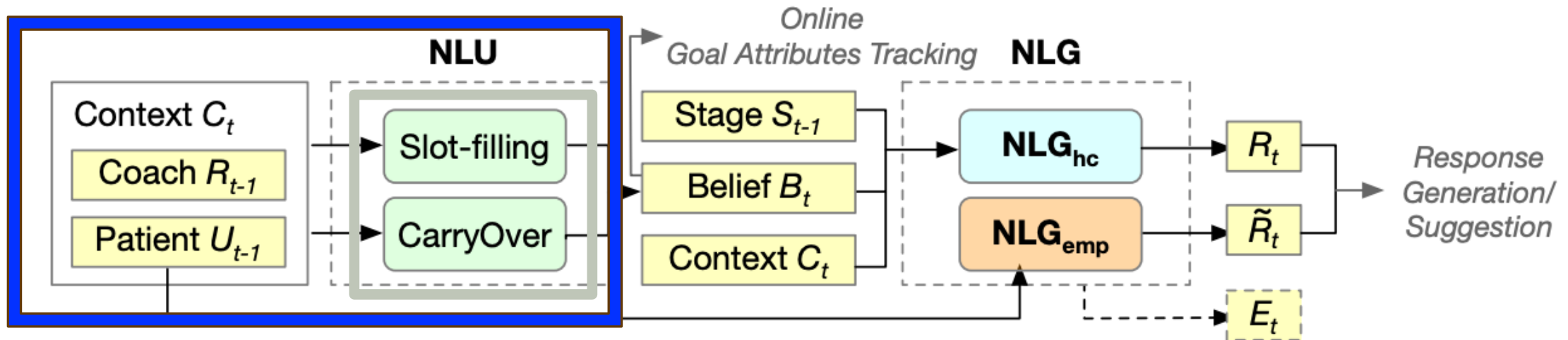
Activity	Time	Location	Amount	Distance	Duration	Days name	Days num	Repetition	Attainability Score
walk	between 6 am to 8 am		4,000 steps	2 miles	30 min	friday saturday		a day	10

- Joint goal accuracy (all 10 attributes correct): 20%
- No response generation

Neural architecture: summarize goal and generate response [Zhou et al COLING 22, COLING-LREC 24]



Neural NLU v1 [Zhou et al COLING 22]



Slot-filling model: jointly extracts slot fillers & intent (dialogue act) from each utterance (inspired by Joint BERT, Chen et al 2019).

Binary Carryover classifier: determines if value of slot should be copied from previous state or updated with new instance seen in U_t

Neural NLU v2: Neurosymbolic AI

... aims to build rich computational AI models, systems and applications by combining neural and symbolic learning and reasoning. It seeks to complement the strengths of deep neural networks and symbolic AI while overcoming their respective weaknesses. <https://nesy-ai.org/>

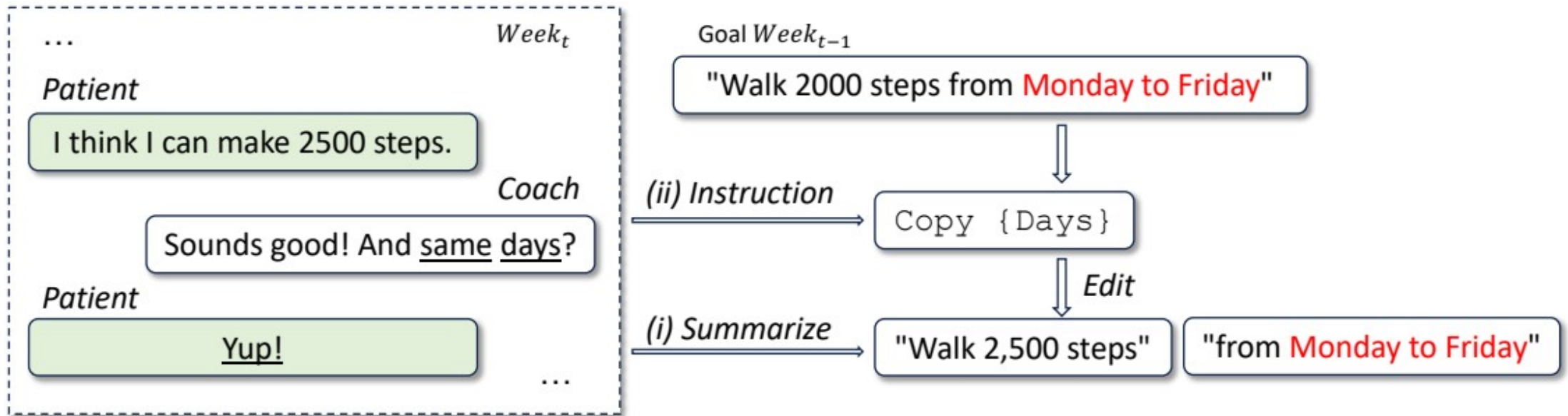
- ❖ Symbolic dialogue managers for Neural Networks [Xompero et al 2020, Zhang et al 2020]
- ❖ Schema-driven control of NN end-to-end [Agrawal et al SigDIAL 24]
- ❖ Deep logic programming approaches to dialogue architectures [Altszyler et al 2021, Zhou et al 2020]



Neurosymbolic AI
[Workshop series \(17 so far\)](#)

Neural NLU v2: Neurosymbolic approach

[Zhou et al LREC-COLING 2024]



End-to-end neuro-symbolic workflow for goal summarization:

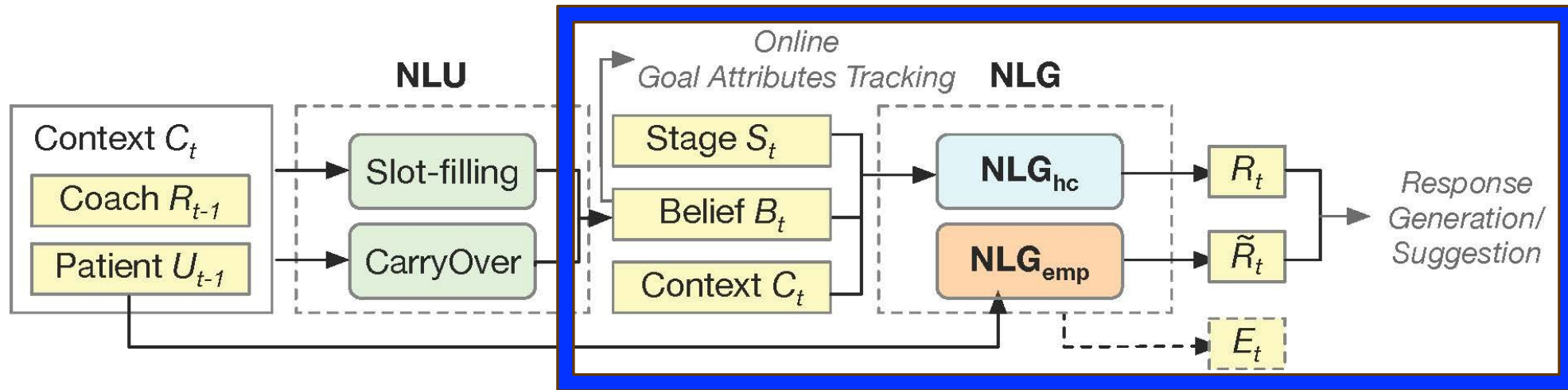
- Neural: model trained to generate *executable* instruction (Copy {Days}) and to extract partial goal ("Walk 2,500 steps")
- Symbolic: Then model edits partial goal on basis of previous goal and instruction

Virtual Coach: NLU Results

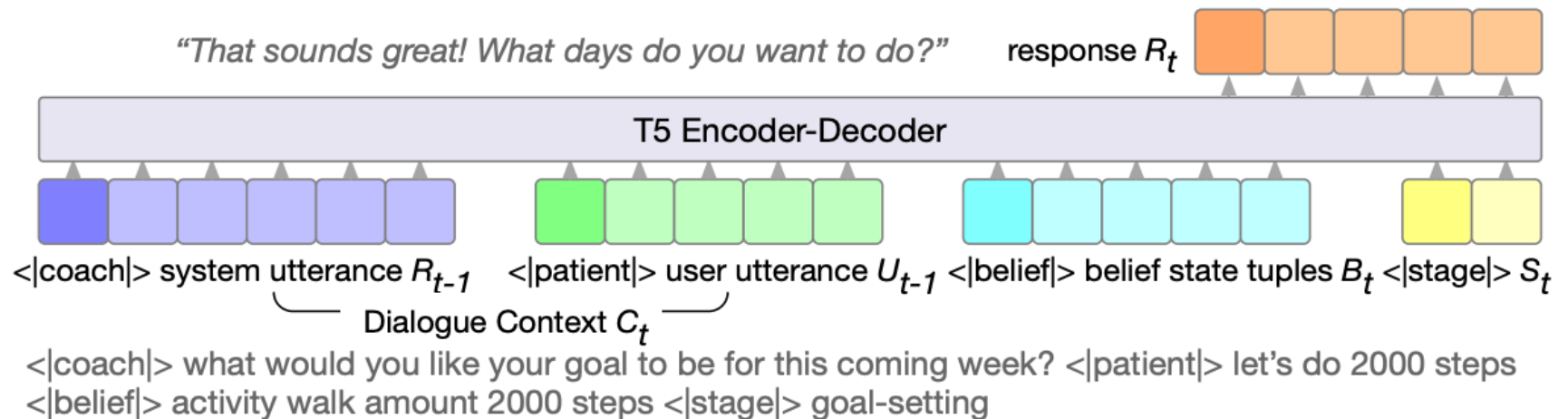
Joint Goal Accuracy (10 attributes all correct):

Classic [Gupta et al SigDIAL 21]	20%
Neural NLU v1 [Zhou et al COLING 22]	21.7%
Neural NLU v2 [Zhou et al LREC-COLING 24]	52.8%

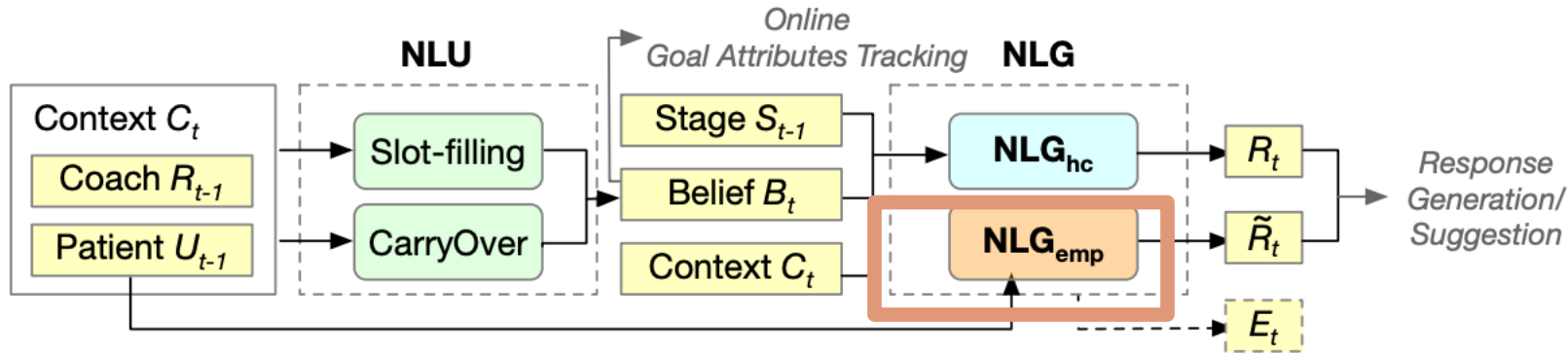
NLG: generate response [Zhou et al COLING 22]



NLG_{hc} : context C_t , belief tokens B_t , and predicted stage S_t embedded with T5 encoder-decoder model [Reffel et al JLMR 2020]



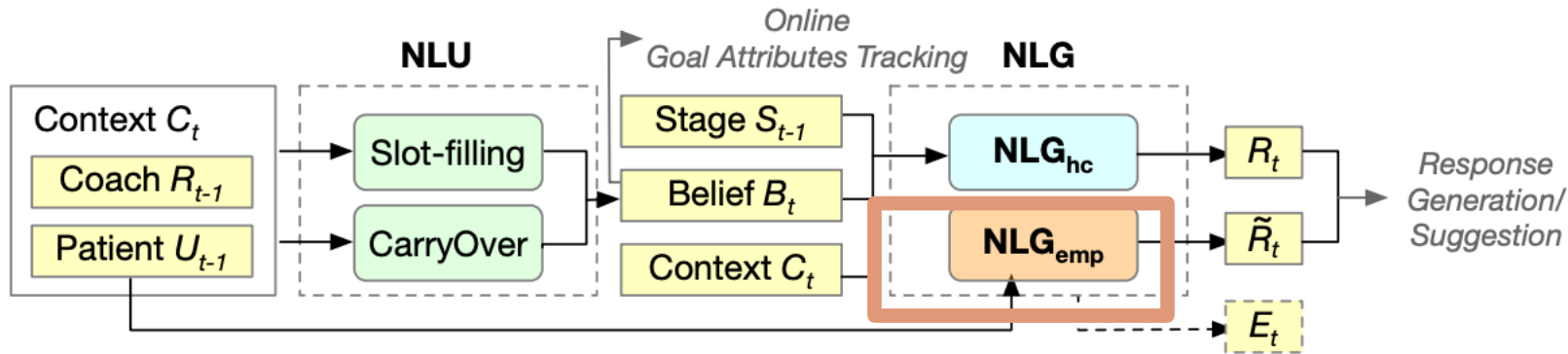
Generate empathetic responses [Zhou et al COLING 22]



Following Sharma et al. (2020), empathetic response based on three communication mechanisms:

- **Emotional Reactions [EMOR]:** “I would be very worried”
- **Interpretations [INTER]:** “I know anxiety is scary”
- **Explorations [EXPL]:** “What happened? How come?”

Generate empathetic responses [Zhou et al COLING 22]



- BERT-based classifier for emotion detection in patient's utterance
- Train GPT2 to generate empathetic responses, given communication mechanism token [EMOR] & patient utterance

<|bos|> [EMOR] I was so exhausted yesterday. <|sep|> That's understandable. Take some rest! <|eos|>
(Patient utterance) (Empathetic response)

Examples of empathetic responses

Coach: Good morning! How is your goal for the week going?

Patient: I am sorry I didn't go to work today I have a massive migraine headache

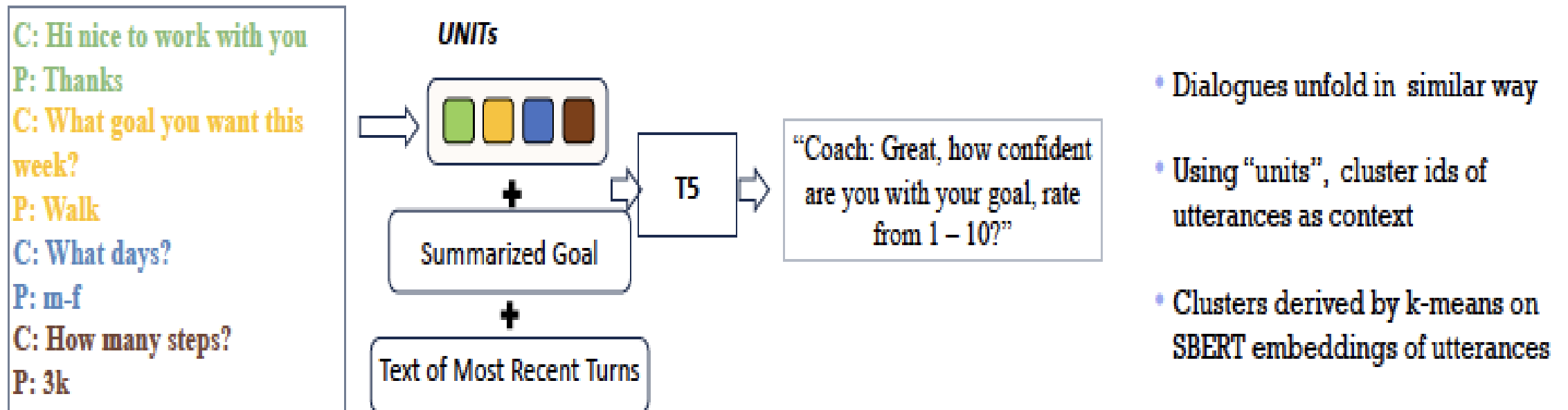
[EMOR] oh no I hope you are OK

[EXPLOR] Oh geez, sorry to hear that. Are you feeling better?

[INTERP] I've had this experience before. Sometimes it really hits you

NLG: Text-Unit-Text generation [Zhou et al LREC-COLING 24]

Text-Unit-Text Dialogue Generation



And what about LLMs? [Zhou et al LREC-COLING 24]

GPT-3.5-turbo: two-shot in-context exemplars due to input context limit

Prompt: As a health coach, your task is to refine the patient's goal into a SMART goal. Ask for details like frequency, time, duration, location, and confidence level, but only one aspect at a time. After establishing the goal, monitor the patient's progress and keep them engaged. Always address their concerns concisely. Now, consider the following dialogue context and formulate your response: ...

Response examples

Coach: Hi [Name], hope you're well. I forgot to message you yesterday to set your goal for Week 6, my apologies!

Coach: You met your goal 5 out of 7 days last week, great job. What would you like to do for this week?

Patient: Gd morning [Name], I am going to set the same goals. With a 7 confidence.

a) Zhou et al. (2022): *That's great. And think about what activity would you like to set for your goal?
Walks, stairs, etc*

b) Zhou et al. (2024): *Ok, great! Can you think of any barriers that might prevent your goal?*

c) GPT-3.5-turbo: *That's fantastic to hear that you're sticking to your goals! Let's make sure it's a SMART goal for Week 6. Can you provide more details about the goal, like the specific number of steps or any changes in frequency, time, or location?*

Human Evaluation

45 nursing students evaluated several contexts + their respective generated responses: about 900 total items, about 20/student

Responses: only [Zhou et al COLING-LREC 24]

- <no edits, minor edits, major edits>
- <irrelevant, harmful, incoherent>

Response Analysis

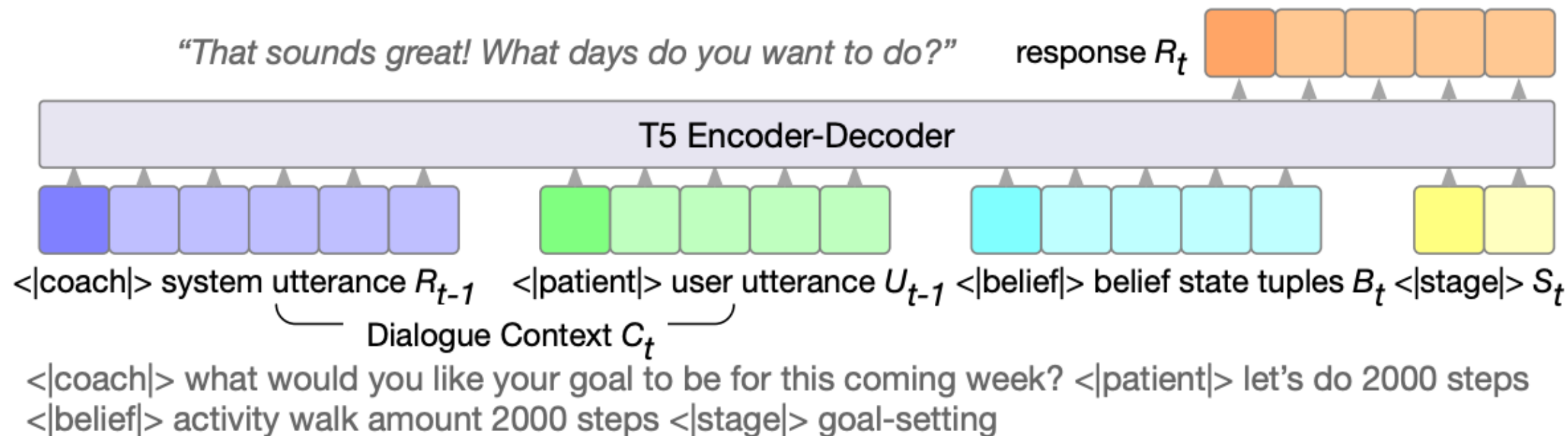
*No harmful or
incoherent
responses*

Virtual Coach: Take aways

- ❖ **Dataset:** collected dataset, partially sharable
- ❖ Developed **neurosymbolic** approach, highest performance
- ❖ **LLMs:** qualitatively, GPT-3.5 answers were incomplete / incoherent
- ❖ **Evaluation:** ran user study with nursing students

Additional slides

NLG [Zhou et al COLING 22]



Context C_t , belief tokens B_t , and predicted stage S_t embedded with T5 encoder-decoder model [Reffel et al JLMR 2020]

References

- I. Gupta, B. Di Eugenio, B. Ziebart, A. Baiju, B. Liu, B. Gerber, L. Sharp, N. Nabulsi, and M. Smart. *Human-human health coaching via text messages: Corpus, annotation, and analysis*. In Proceedings of the 21th Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 246–256, 1st virtual meeting, July 2020. Association for Computational Linguistics.
- I. Gupta, B. Di Eugenio, B. Ziebart, B. Liu, B. Gerber, and L. Sharp. *Goal summarization for human-human health coaching dialogues*. In The Thirty-Third International FLAIRS Conference, 2020.
- I. Gupta, B. Di Eugenio, B. D. Ziebart, B. Liu, B. S. Gerber, and L. K. Sharp. *Summarizing behavioral change goals from SMS exchanges to support health coaches*. In Proceedings of the 22nd Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 276–289, 2021.
- Gupta, Itika, Barbara Di Eugenio, Devika Salunke, Andrew Boyd, Paula Allen-Meares, Carolyn Dickens, and Olga Garcia. "Heart failure education of African American and Hispanic/Latino patients: Data collection and analysis." In *Proceedings of the First Workshop on Natural Language Processing for Medical Conversations*, pp. 41-46. 2020.
- Mac, O., et al . (2022). Comparison of Readability Scores for Written Health Information Across Formulas Using Automated vs Manual Measures. *JAMA Network Open*, 5(12)
- C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. *Exploring the limits of transfer learning with a unified text-to-text transformer*. Journal of Machine Learning Research, 21(140):1–67, 2020.
- Su, Y., Shu, L., Mansimov, E., Gupta, A., Cai, D., Lai, Y.-A., and Zhang, Y.: *Multi-task pre-training for plug-and-play task-oriented dialogue system*. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), eds. S. Muresan, P. Nakov, and A. Villavicencio, pages 4661–4676, Dublin, Ireland, May 2022. Association for Computational Linguistics.

References (ctd)

- Tayal, Anuja, Barbara Di Eugenio, Devika Salunke, Andrew Boyd, Carolyn A. Dickens, Eulalia P. Abril, Olga Garcia-Bedoya, and Paula G. Allen-Meares. "A neuro-symbolic approach to monitoring salt content in food." In *Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health) @ LREC-COLING 2024*, pp. 93-103. 2024.
- Y. Zhou, B. Di Eugenio, B. Ziebart, L. Sharp, B. Liu, B. Gerber, N. Agadacos, and S. Yadav. Towards enhancing health coaching dialogue in low-resource settings. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 694–706, Gyeongju, Republic of Korea, Oct. 2022. International Committee on Computational Linguistics.
- Y. Zhou, B. Di Eugenio, B. Ziebart, L. Sharp, B. Liu, and N. Agadacos. Modeling lowresource health coaching dialogues via neuro-symbolic goal summarization and text-units text generation. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 11498–11509, Torino, Italia, May 2024. ELRA and ICCL.
- Y. Zhou, B. Di Eugenio, and L. Cheng. *Unveiling performance challenges of large language models in low-resource healthcare: A demographic fairness perspective*. In O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, and S. Schockaert, editors, *Proceedings of the 31st International Conference on Computational Linguistics*, pages 7266–7278, Abu Dhabi, UAE, Jan. 2025. Association for Computational Linguistics.
- Y. Zhou and B. Di Eugenio. *Veracity bias and beyond: Uncovering LLMs' hidden beliefs in problem-solving reasoning*. In W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21298–21310, Vienna, Austria, July 2025. Association for Computational Linguistics.